

# GAC 135 (UNIT 4)

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## The Nature of Probability and Statistics

- **Statistics** is the science of conducting studies to collect, organize, summarize, analyze, and draw conclusions from data.
- A **variable** is a characteristic or attribute that can assume different values.
- The word **data** means facts or information collected for reference or analysis.
- **Descriptive** consist of the collection, organization, summarization, and presentation of data.
- **Inferential statistics** consist of generalizing from samples to populations, performing estimations and hypothesis tests, determining relationships among variables, and making predictions.

- A **population** consists of all subjects (human or otherwise) that are being studied.
- A **sample** is a group of subjects selected from a population.
- A **statistic** is a characteristic or measure obtained by using the data values from a sample.
- A **parameter** is a characteristic or measure obtained by using all the data values from a specific population.

## TYPES OF DATA AND STATISTICAL DIAGRAMS

Data may be classified into various types. Firstly, we can divide data into two groups, namely

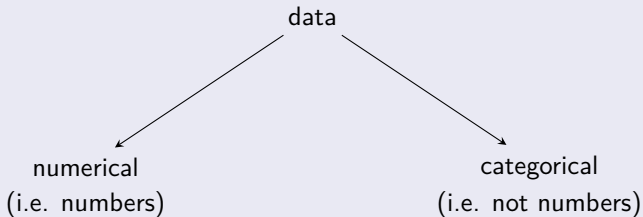


Figure 1

- **Numerical data (quantitative data)** values are numbers or quantities (e.g 3.5 years, 1.2 kg, 4 ms).
- **Categorical data** (qualitative data) values are not numerical in nature (e.g sex, eye colour, type of policy) but relate to categories (e.g male/female, blue/green/brown/..., whole life assurance/term assurance/...).

### Example 1.1

State whether the following data values are numerical or categorical.

- 1) weight
- 2) place of birth
- 3) type of paint used on a vehicle
- 4) amount of voltage in a light bulb
- 5) age
- 6) amount of oil in a car engine

### Example 1.2

State whether the following types of numerical data are discrete or continuous

- 1) weight
- 2) Temperature
- 3) age
- 4) Time taken to complete a task

We can further subdivide numerical data into two types:

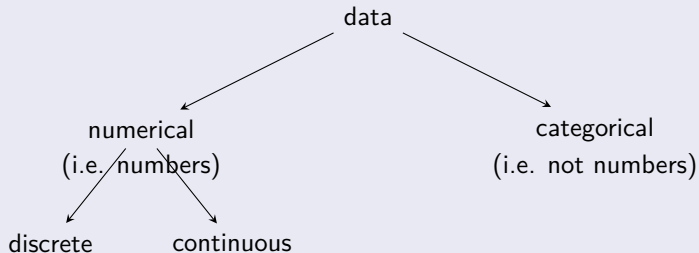


Figure 2



- **Attribute** (or dichotomous) data values have only two categories, e.g., claim/no claim, dead/alive or male/female.
- **Nominal data values** cannot be ordered in any natural way, e.g., type of policy (whole life assurance, term assurance or endowment assurance) or nature of claim (fire, theft, accident, earthquake, etc.).
- **Ordinal data values** can be ordered in a natural way, e.g., exam grades (A, B, C, ...), level of agreement (strongly agree, agree, neutral, disagree, strongly disagree).

# Summarising data in tables

## Frequency distributions

Suppose that the number of vehicles fixed each day to a small mechanic shop over the last 28 working days are as follows:

4, 2, 0, 3, 2, 1, 1, 4, 2, 5, 0, 3, 2, 1, 3, 4, 3, 5, 1, 2, 4, 2, 3, 1, 4, 2, 3, 2

Table 1.1

| Vehicles fixed each day | Frequency |
|-------------------------|-----------|
| 0                       | 2         |
| 1                       | 5         |
| 2                       | 8         |
| 3                       | 6         |
| 4                       | 5         |
| 5                       | 2         |

**Table 1.1** is called a frequency distribution. It shows the number of times that each value has been observed.  
A frequency distribution is suitable for categorical or discrete numerical data.

### Example 1.11

Explain how we can obtain the following from the above frequency distribution table:

- 1) the total number of observations (data items)
- 2) the total number of vehicles fixed each day.

## Grouped frequency distributions

Below is a list of the ages of retired mechanical engineers in XYZ company:

57 68 75 66 72 86 80 81 70 78 76 72 88 84 69

77 83 90 48 63 74 81 94 51 73 96 81 66 77 101

This data can be summarised using the grouped frequency distribution.

## Cumulative frequency

**Cumulative frequency** is the sum of all the frequencies up to and including the current point.

# Summarising data in diagrams

## Bar charts

A **bar chart** can be drawn for discrete numerical data or for categorical data. For each observed value, we draw a bar to show the number of times that the value has occurred (i.e. its frequency).

## Histogram

A **histogram** is similar to a bar chart but there are a few important differences .



A histogram is used to display continuous data. We therefore use a continuous scale with no spaces between the bars. In addition, we use frequency density rather than frequency on the vertical axis.

**Frequency density** Frequency density is defined as follows.

$$\text{frequency density} = \frac{\text{frequency}}{\text{class width}}$$

Class width is the difference between the largest possible value in the group (or class) and the smallest. By setting the height of each bar equal to its frequency density, the area of each bar is equal to its frequency.

# Cumulative frequency curves

## Constructing a cumulative frequency curve

To obtain the curve, we plot a graph of the cumulative frequencies against the upper limit of each class and join the points.

## Location

The **location** of a data set tells us about the typical size of the observations. One way to compare locations is through the use of stem and leaf diagrams.

## Spread

We can easily compare the spreads of two or more data sets using dotplots.

## Skewness(Shape)

The **skewness** tells us how symmetrical a data set is — the greater its asymmetry, the greater the magnitude of its skewness.

positively skewed



symmetric



negatively skewed



Figure 4

## Sample Mode

The mode of a data set or sample is the value with the highest frequency, i.e. the value that occurs most often. The mode is also sometimes referred to as the modal value.

### Example 1.13

Below are the numbers of new employees taken on in the last year by six companies:

8   5   19   3   6   5

Identify the modal value.

### Example 1.14

Identify the mode of each of the following data sets:

- 6 4 7 5 4 6
- 1 2 3 4 5



## MODALS GROUP

For the grouped data, the modal group with the highest frequency.

### Example 1.15

The grouped frequency distribution below summarises the ages of retired mechanical engineers in XYZ company.

| Ages    | Frequency |
|---------|-----------|
| 40-49   | 2         |
| 50-59   | 5         |
| 70-79   | 10        |
| 88-89   | 3         |
| 90-99   | 8         |
| 100-109 | 1         |

Identify the modal group.

## Advantages

- The sample mode is easy to calculate.
- The sample mode is not affected by extreme values in the data set.

## Disadvantages

- The sample mode may not be unique.
- The sample mode may not exist.
- The sample mode focuses on a small number of data values.
- The sample mode does not have a simple algebraic formula and, as a result, it is not useful in further calculations.

## Sample mean

The **mean** of a data set is obtained by summing the observations and dividing the total by the number of observations in the sample. The sample mean of the data set containing the observations  $x_1, x_2, \dots, x_n$  is given by:

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{1}{n} \sum_{i=1}^n x_i$$

### Example 1.16

The sizes of the last ten car insurance claims paid by an insurance company are:

|           |           |           |           |            |
|-----------|-----------|-----------|-----------|------------|
| GHS 1,500 | GHS 1,820 | GHS 840   | GHS 260   | GHS 2,100  |
| GHS 790   | GHS 530   | GHS 1,360 | GHS 1,780 | GHS 1,650. |

Determine the mean of these observations.

Below are the salaries (in GHS 000s) of 8 individuals in a small company:

25   25   25   25   25   25   25   50

Calculate the mean salary for this sample.

### Calculating the sample mean from a frequency distribution

$$\bar{x} = \frac{x_1 f_1 + x_2 f_2 + \cdots + x_m f_m}{f_1 + f_2 + \cdots + f_m} = \frac{\sum_{i=1}^m x_i f_i}{\sum_{i=1}^m f_i}$$

### Example 1.17

The grouped frequency distribution below summarises the ages of 30 retired engineers in XYZ company.

| Ages      | 40–49 | 50–59 | 60–69 | 70–79 | 80–89 | 90–99 | 100–109 |
|-----------|-------|-------|-------|-------|-------|-------|---------|
| Frequency | 1     | 2     | 5     | 10    | 8     | 3     | 1       |

Estimate the mean of the exact ages of this sample of retired engineers.



## Advantages

- The sample mean uses all the data values.
- The sample mean has mathematical properties that make it useful in further calculations.

## Disadvantages

- The sample mean can be distorted by extreme values.

## Sample Median

The **median** of a data set is the value that splits the set into two equal halves, so that half of the observations are below the median and half are above it. When the exact value of each observation is known, the median is the middle value in the set when the observations are ordered from smallest to largest.

### Example 1.18

Determine the median of the following observations:

125   75   25   20   50   25   50   15   30

### Example 1.19

Determine the median of the following data set:

9 1 4 10 15 5 9

## Position of the median value in a list of ordered observations

If we have a list of  $n$  observations, arranged in order from smallest to largest, then the position of the median value is

$$\frac{n + 1}{2}$$

If  $n$  is an even number, the median is defined to be the average of the middle two values, ie the average of the  $\frac{n}{2}th$  ( $\frac{n}{2} + 1$ ) the values.

### Example 1.20

Below are the salaries (in GHS 000s) of 8 individuals in a small company:

25 25 25 25 25 25 25 50

Determine the median salary for this sample.

# Estimating the sample median using a grouped frequency distribution

## Estimating the Median

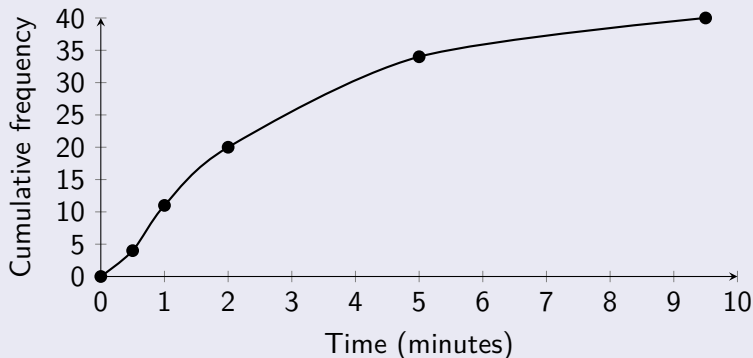
If we have a grouped frequency distribution involving a total of  $n$  observations, we:

- identify the interval in which the cumulative frequency reaches  $\frac{n}{2}$
- use linear interpolation to estimate the median value

# Estimating the sample median using a cumulative frequency diagram

## Example 1.21

A consumer watchdog measures the length of time (in minutes) for which 40 telephone calls to a helpline were put on hold. A cumulative frequency graph of the data is given below.





## Advantages

- The sample median is not affected by extreme values.

## Disadvantages

- The sample median does not use all the data values directly.
- The sample median does not have a simple algebraic formula and, as a result, it is not useful for further calculations.

# Transforming Data Sets

## Measures of location for a transformed data set

If each value in a data set is increased by a constant amount,  $a$  (which might be negative), then the sample mode, mean and median are also each increased by  $a$ .

If each value in a data set is multiplied by a constant,  $b$ , then the sample mode, mean and median are equal to their original values each multiplied by  $b$ .

# Shape of a data set

## Relative positions of mode, median and mean

The relative positions of the mode, median and mean tell us about the shape of a data set.

- $\text{mode} = \text{median} = \text{mean}$ , for a symmetrical data set
- $\text{mode} < \text{median} < \text{mean}$ , for a positively skewed data set
- $\text{mode} > \text{median} > \text{mean}$ , for a negatively skewed data set

# Measures of Spread and Skewness

## Range of a data set

The **range** of a data set is the difference between its largest value and its smallest value.

### Example 1.22

Calculate the range of the following data set:

8 5 19 3 6 5

### Example 1.23

Below are the salaries of the employees of a small company.

| Salary (GHS) | Frequency |
|--------------|-----------|
| 25,000       | 7         |
| 50,000       | 1         |

Calculate the range of this data set.

## Advantage

- The range is very easy to calculate

## Disadvantages

- The range is affected by extreme values.
- The range focuses on a small number of data values.
- The range does not have convenient mathematical properties and as a result is, not useful

# Interquartile Range

## Lower and upper quartiles

The **lower quartile** of a data set is the point that is one quarter of the way through the data set when the values are arranged in increasing order. It is denoted by the symbol  $Q_1$ .

The **upper quartile** of a data set is the point that is three-quarters of the way through the data set when the values are arranged in increasing order. It is denoted by the symbol  $Q_3$ .



## Interquartile range

The **interquartile range** of a data set is:

$$Q_3 - Q_1$$

If the values in a data set are spread out, the upper and lower quartiles will usually be a long way apart and the interquartile range will have a high value.

## Positions of the lower and upper quartiles

For a data set that contains  $n$  observations, arranged in order from smallest to largest, the position of the lower quartile is:

$$\frac{n+1}{4}$$

and the position of the upper quartile is:

$$3\left(\frac{n+1}{4}\right)$$

### Example 1.24

Identify the lower quartile, median, upper quartile and interquartile range of the following data set:

GHS 50 GHS120 GHS 40 GHS 30 GHS 15 GHS50 GHS20

### Example 1.25

Calculate the lower quartile, upper quartile and interquartile range of the data set

3   5   6   9   11   16

### Example 1.26

Calculate the lower quartile, upper quartile and interquartile range of the data set

|           |   |   |   |   |   |
|-----------|---|---|---|---|---|
| Values    | 1 | 2 | 3 | 4 | 5 |
| Frequency | 3 | 4 | 6 | 5 | 2 |

## Advantages

- The interquartile range is not affected by extreme values of a data set.
- The interquartile range uses more data values than the range.

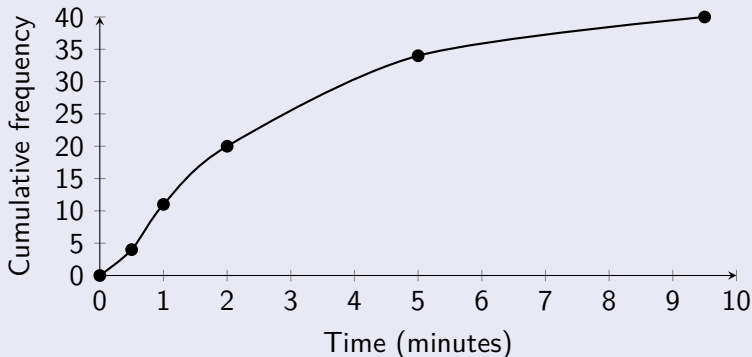
## Disadvantages

- The interquartile range does not use all the data values.
- The interquartile range does not have convenient mathematical properties and, as a result, it is not useful for further calculations.

# Estimating the quartiles using a cumulative frequency diagram

## Example 1.27

A consumer watchdog measures the length of time (in minutes) for which 40 telephone calls to a helpline were put on hold. A cumulative frequency graph of the data is given below.



# Sample variance and standard deviation

## Formulae for sample variance and sample standard deviation

The **sample variance** of the data set containing the observations  $x_1, x_2, \dots, x_n$  is given by:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

The **sample standard deviation** of the data set containing the observations  $x_1, x_2, \dots, x_n$  is given by:

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$



## Alternative formulae for sample variance and sample standard deviation

The **sample variance** of the data set containing the observations  $x_1, x_2, \dots, x_n$  is given by:

$$s^2 = \frac{1}{n-1} \left( \sum_{i=1}^n x_i^2 - n\bar{x}^2 \right)$$

The **sample standard deviation** of the data set containing the observations  $x_1, x_2, \dots, x_n$  is given by:

$$s = \sqrt{\frac{1}{n-1} \left( \sum_{i=1}^n x_i^2 - n\bar{x}^2 \right)}$$

### Example 1.28

Below are the salaries (in GHS 000s) of eight individuals in a small company:

25   25   25   25   25   25   25   50

Calculate the standard deviation of the salaries in this sample.

# Calculating the sample variance and sample standard deviation from a frequency distribution

## Calculating the sample variance from a frequency distribution

If we have a sample that contains  $m$  different values,  $x_1, x_2, \dots, x_m$ , with frequencies  $f_1, f_2, \dots, f_m$  respectively, then the sample variance of these observations is given by:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^m (x_i - \bar{x})^2 f_i = \frac{1}{n-1} \left( \sum_{i=1}^m x_i^2 f_i - n\bar{x}^2 \right)$$

Where:

$$n = \sum_{i=1}^m f_i$$

and

$$\bar{x} = \frac{\sum_{i=0}^m x_i f_i}{\sum_{i=0}^m f_i} = \frac{\sum_{i=0}^m x_i f_i}{n}$$

### Example 1.29

The frequency table below shows the number of times a light bulb was changed in the last year in an office:

| Number of times a light bulb was changed | Frequency |
|------------------------------------------|-----------|
| 0                                        | 74        |
| 1                                        | 19        |
| 2                                        | 5         |
| 3                                        | 2         |

Calculate the sample standard deviation of the number of times a light bulb was changed.

### Example 1.30

The grouped frequency distribution below summarises the ages of 30 retired engineers in XYZ company.

| <b>Ages</b> | <b>Frequency</b> |
|-------------|------------------|
| 40–49       | 1                |
| 50–59       | 2                |
| 60–69       | 5                |
| 70–79       | 10               |
| 80–89       | 8                |
| 90–99       | 3                |
| 100–109     | 1                |

Estimate the sample standard deviation of the exact ages of retired engineers.

## Advantages

- The sample variance and sample standard deviation use all the values in a data set.
- The sample variance has mathematical properties that make it useful in further calculations.
- The sample standard deviation has the same units as the data values.

## Disadvantages

- The sample variance and sample standard deviation are affected by extreme values (but to a lesser extent than the range).

# Transforming Data Sets

## Measures of spread and skewness for a transformed data set

If each value in a data set is transformed by adding a constant,  $a$  (which might be negative), then the range, interquartile range, standard deviation, variance and third central moment are unaffected.

If each value in a data set is multiplied by a constant,  $b$  then:

- the range, interquartile range and standard deviation are also multiplied by a factor of  $b$ ;
- the variance is multiplied by a factor of  $b^2$
- the third central moment (skewness) is multiplied by a factor of  $b^3$ .



## 2.0 Probability

### Basic Probability

**Probability theory** is used to explain and predict the outcomes of future uncertain events. Examples in actuarial science include the uncertainties involved in finance and insurance.

### Example 2.1

A 10p and a 50p coin are both tossed, and their outcomes (head or tail) recorded. List all the possible outcomes (i.e. give the sample space).

## Basic properties satisfied by the probability of occurrence of an event

- 1  $0 \leq P(A) \leq 1$
- 2  $P(A) = 0$  if  $A$  is an impossible event
- 3  $P(A) = 1$  if  $A$  is a certain event

## Probability of an event occurring

The probability that event  $A$  occurs is given by:

$$P(A) = \frac{\text{number of equally likely outcomes in which event } A \text{ can happen}}{\text{total number of equally likely outcomes}}$$

### Example 2.2

A die is rolled. Calculate the probability that outcome is a number greater than 4.

### Example 2.3

A pile of 15 examination scripts contains 2 Mathematics (Pure), 3 Mathematics (Applied), 4 English and 6 Chemistry papers. A marker selects a script from the pile at random.

Calculate the probability that the selected script is for a Mathematics subject.

### Probability of a complementary event

$$P(A') = P(\overline{A}) = 1 - P(A)$$

where  $A'$  (or  $\overline{A}$ ) is the complementary event of event  $A$ .

### Example 2.4

The probability that salary payments exceeds GHS 1,000 is 0.6.  
Calculate the probability that the salary payment does not exceed GHS 1,000.



### Addition rule for mutually exclusive events

For any two mutually exclusive events  $A$  and  $B$

$$P(A \text{ or } B) = P(A) + P(B)$$

### Example 2.6

Consider again the situation described in Example 2.3, where a pile of 15 exam scripts contains 2 Mathematics (Pure), 3 Mathematics (Applied), 4 English and 6 Chemistry papers, and a script is selected from the pile at random. Calculate the probability that the selected script is for any mathematics or science subject.

## Example 2.7

In a traffic survey the probabilities of observing various types of cars are given below:

| Feature | Probability |
|---------|-------------|
| blue    | 0.25        |
| white   | 0.3         |
| silver  | 0.15        |
| Ford    | 0.3         |
| Renault | 0.2         |

- 1) Calculate the probability that the next car:
  - 1) is a Ford or a Renault
  - 2) is blue, white or silver
- 2) Explain why the probability that the next car is a Ford or blue is not equal to  $0.3 + 0.25$ .

# Addition rule for non-mutually exclusive events

## Addition rule

For any two events  $A$  and  $B$ :

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The addition rule for mutually exclusive events is a special version of this more general rule. This is because if events  $A$  and  $B$  are mutually exclusive then:

$$P(A \text{ and } B) = P(A \cap B) = 0$$

## Example 2.8

In a group of students it is known that 45% watch Alias, 50% watch West Wing and 30% watch Alias and West Wing on television. Calculate the probability that a student watches Alias or West Wing.

# Multiplication rule for independent events

We are now going to calculate the probability of two events both happening, i.e. a probability of the form  $P(A \text{ and } B)$ , where the two events are independent. For any two independent events  $A$  and  $B$ :

$$P(A \text{ and } B) = P(A) \times P(B)$$

### Example 2.9

State, with brief reasons, which of the following pairs of events are independent:

- 1) arriving in time to catch the early bus and getting to work on time
- 2) one policyholder dying and a second policyholder dying
- 3) rolling a 6 on one die and rolling a 6 on a second die
- 4) Team A winning a football match and Team B winning the same football match

### Example 2.10

A coin is tossed and a die is rolled. Calculate the probability of the coin landing tails and obtaining a number greater than 4 on the die.



### Example 2.11

A gaming machine has two wheels. Each wheel spins and then stops to reveal a picture. The probability that the picture on the first wheel is a cherry is  $\frac{3}{10}$ , whereas for the second wheel the probability is  $\frac{2}{5}$ . The two wheels operate independently.

Calculate the probability of obtaining:

- 1 a cherry on the first wheel and on the second wheel
- 2 a cherry on the second wheel but not on the first wheel
- 3 no cherries on either of the two wheels

## Multiplication rule for dependent events

For any two events  $A$  and  $B$ :

$$P(A \text{ and } B) = P(A) \times P(B | A)$$

### Example 2.12

A regular theatre-goer likes to sit in a particular seat in the auditorium. If she books her theatre ticket early, the probability that she gets her favourite seat is 0.8. If she books her theatre ticket late, the probability that she gets her favourite seat is 0.3. Write down the following probabilities:

- $P(\text{gets favourite seat} \mid \text{early booking})$
- $P(\text{does not get favourite seat} \mid \text{late booking})$

### Example 2.13

There are 3 jam doughnuts and 4 apple doughnuts for sale in a baker's shop, but the baker cannot remember which is which. They all look the same from the outside. A customer buys a doughnut, and leaves the shop before eating it. Later, another customer arrives and wants to buy a second doughnut. Find:

- 1  $P(\text{2nd doughnut is apple} \mid \text{1st doughnut was jam})$
- 2  $P(\text{2nd doughnut is apple} \mid \text{1st doughnut was apple})$

### Example 2.14

William's sock drawer contains 4 blue socks and 4 black socks.

- ① William takes two socks from the drawer at random. Calculate the probability that:
  - he chooses a blue sock followed by a black sock.
  - he chooses a pair of black socks.
- ② To be sure that he gets a matching pair of socks, William decides to take three socks (chosen at random) from the drawer. Calculate the probability that:
  - he chooses 3 blue socks.
  - he chooses a blue sock followed by 2 black socks.

# Unconditional Probability of a Dependent Event

## Unconditional probability of a dependent event occurring

For an event  $B$ , which is dependent on mutually exclusive and exhaustive events  $A_1, A_2, \dots, A_n$ :

$$P(B) = P(A_1) \times P(B | A_1) + P(A_2) \times P(B | A_2) + \dots + P(A_n) \times P(B | A_n)$$

### Example 2.15

Kate catches the bus to work. The probability that the bus is late on any day is 0.6. If the bus is late, the probability that Kate is late for work is 0.8, otherwise it is 0.3. Calculate:

- the probability that the bus is late and Kate is late for work
- the probability that the bus is early and Kate is late for work
- the probability that Kate is late for work

## 3.0 Permutations and Combinations

### Simple Choosing

If we have to make  $r$  different choices and there are  $n_1$  options for the first choice,  $n_2$  options for the second choice, and so on, then the total number of different choices that can be made is:

$$n_1 \times n_2 \times \cdots \times n_r$$



### Example 3.1

At a restaurant the menu has:

- 2 starters (soup or prawn cocktail)
- 4 main courses (chicken, pasta, fish or risotto)
- 3 desserts (chocolate cake, crepes or ice cream)

Calculate the number of different meals that can be ordered.

# Permutations of all objects in a group

The number of ways of ordering  $n$  different objects is:

$$n! = n \times (n - 1) \times \cdots \times 3 \times 2 \times 1$$

This factorial function is also available on most calculators.

## Example 3.2

I need to put the following coins into a meter: GHS 2, GHS 1, 50p, 20p, 10p. Calculate the number of different orders in which this can be done.

## Permutations of some of the objects selected in a group

The number of ways in which  $r$  objects, chosen from  $n$  different objects, can be ordered is:

$${}^n P_r = \frac{n!}{(n-r)!}$$

### Example 3.3

Thirty people buy one raffle ticket each, to win one of four prizes. The prizes are a car, a holiday, a bicycle and some wine. Calculate the number of different ways in which the prizes can be allocated.

# Combinations

The number of ways of choosing  $r$  objects from  $n$  different objects, when the order is not important, is:

$${}_nC_r = \binom{n}{r} = \frac{n!}{(n-r)! r!}$$

## Example 3.4

A football team manager has a squad of 20 players and needs to pick a team of 11 for Saturday's game. Calculate the number of different teams that can be selected.

## Example 3.5

In a lottery described in the example above, a lottery ticket that includes exactly 3 of the numbers selected by the machine will win its owner GHS 10.

- 1 Calculate the number of ways of matching exactly 3 numbers chosen by the machine, when a person chooses 6 numbers from 49.
- 2 Calculate the probability that a person who buys one lottery ticket will win a GHS 10 prize.

**Hint:** You will need to use your answer from the example above.

### Example 3.6

In the example above, a male student (Robert) and a female student (Aaisha) are both keen to be selected in the student research group. Assuming that students are selected from the 10 male and 8 female students at random, calculate the probability that both Robert and Aaisha are chosen for the research group.

# Permutations of all objects when some are identical

The number of distinct orderings of  $n$  objects of which  $n_1$  are identical and of Type 1,  $n_2$  are identical and of Type 2,  $\dots$ ,  $n_r$  are identical and are of Type  $r$  is equal to the multinomial coefficient:

$$\frac{n!}{n_1! \times n_2! \times \cdots \times n_r!}$$

The number of permutations of a group of  $n$  objects of which  $r$  objects are of Type 1 and  $n - r$  objects are of Type 2, is:

$$\binom{n}{r} = \frac{n!}{(n-r)! r!}$$



## Exercises

- 1) The distances travelled by snails in 5 minutes were measured to the nearest centimetre. The results are shown in the table below:

| Distance (cm) | Frequency |
|---------------|-----------|
| 0–4           | 4         |
| 5–6           | 7         |
| 7–8           | 15        |
| 9–12          | 23        |
| 13–18         | 11        |

Calculate the frequency densities that you would use to draw a histogram of this data set.

- 2) The mortality of males before the state retirement age is being investigated. The ages last birthday at death of 500 males were recorded and are summarised in the table below:

| <b>Age</b>       | 5–19 | 20–29 | 30–39 | 40–49 | 50–54 | 55–59 | 60–64 |
|------------------|------|-------|-------|-------|-------|-------|-------|
| <b>Frequency</b> | 3    | 20    | 27    | 63    | 67    | 116   | 204   |

- (i) Draw the histogram to represent these data values.

- 3) The following data values are the total number of hours spent processing some insurance claims:

8.02 5.11 5.04 3.88 4.76 3.26 4.41 5.19 4.48 6.28  
9.12 6.53 5.14 2.57 6.80 7.31 5.71 6.16 7.51 8.58

- (i) Display this data set in a stem and leaf diagram by rounding to 1 decimal place.
- (ii) Comment on the shape of the distribution.

- 4) The length of time (in minutes) for which calls to a helpline were put on hold are given in the following table:

| Time ( $t$ )     | Frequency |
|------------------|-----------|
| $0 < t \leq 0.5$ | 2         |
| $0.5 < t \leq 1$ | 5         |
| $1 < t \leq 2$   | 7         |
| $2 < t \leq 5$   | 12        |
| $5 < t \leq 10$  | 4         |

- (i) Construct a cumulative frequency curve for this data set.
- (ii) Use this graph to estimate:
- (a) how many calls were held for less than 3 minutes;
  - (b) the time for which more than 50% of the calls were on hold.

- 5) The following table gives the ages of 100 men (in years) in the form of a grouped frequency distribution:

| Age last birthday | Number of men |
|-------------------|---------------|
| 20 – 24           | 1             |
| 25 – 29           | 2             |
| 30 – 34           | 10            |
| 35 – 39           | 16            |
| 40 – 44           | 22            |
| 45 – 49           | 20            |
| 50 – 54           | 15            |
| 55 – 64           | 14            |

Draw a histogram of the data.

6)

| Claim size (GHS)       | Frequency |
|------------------------|-----------|
| $\leq 100$             | 862       |
| $> 100$ and $\leq 200$ | 608       |
| $> 200$ and $\leq 300$ | 1,253     |
| $> 300$ and $\leq 400$ | 1,066     |
| $> 400$ and $\leq 500$ | 558       |
| $> 500$                | 1,290     |
| Total                  | 5,637     |

- (i) Calculate the cumulative frequencies and draw a graph of the claim size distribution function.
- (ii) Estimate the proportion of claim sizes that are less than GHS 250.

7) The ages of employees in two departments are given below:

|           |                                        |
|-----------|----------------------------------------|
| Marketing | 24, 25, 27, 27, 28, 28, 28, 29, 29, 32 |
| Personnel | 27, 31, 35, 38, 44, 44, 47, 47, 47, 51 |

Draw dotplots for each department and use them to compare the ages of their employees.